Allison Welch

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SUMMARY

- 5+ years experience in **geospatial analysis**, satellite remote sensing, GIS, and **emissions modeling** with expertise in wildfire emissions quantification and air quality monitoring methodologies. Proficient in experimental design, field operations, sampling methodologies for criteria pollutant quantification, analytical chemistry analysis, and **developing workflow using program languages**. Skilled in database development, **reconciling datasets** across space and time, quality assessment techniques, and managing environmental data using geospatial software.
- Action-oriented, environmental outcomes-focused researcher with demonstrated expertise in wildfire impacts on atmospheric chemistry and air quality-climate interactions. Experienced in stakeholder engagement, technical writing for grants and publications, and translating complex scientific findings for technical and non-technical audiences, evidenced by 7 peer-reviewed publications (2 first-authored) in high impact journals including *JGR: Atmospheres*, *Nature Communications*, and *PNAS* and 14+ conference presentations including 3 first-authored at American Geophysical Union meetings.
- Strong leadership and mentoring experience with proven ability to manage cross-functional research teams, secure external funding ($50,000 NSF award), and collaborate with multidisciplinary scientists to inform air quality improvement strategies and emission reduction solutions.

EDUCATION

University of California, Irvine, Irvine, CA — June 2025
Ph.D., Earth System Science
Dissertation: *Changing Arctic Vegetation: Examining Shrubification, Permafrost Vulnerability, and Land-Atmosphere Interactions*

University of California, Irvine, Irvine, CA — June 2022
M.S., Earth System Science

University of Florida, Gainesville, FL — May 2020
B.S., Environmental Science, Summa cum Laude

TECHNICAL SKILLS

- Programming Languages: Proficient in Python and R for data processing, analysis, and visualization; intermediate in JavaScript for Google Earth Engine, SQL
- Geospatial & Remote Sensing: ArcGIS, QGIS, Google Earth Engine, satellite remote sensing, drone remote sensing, spatial data analysis, image segmentation
- Atmospheric & Environmental Modeling: MEGAN (biogenic emissions), HYSPLIT (transport modeling), numerical and biogeochemical modeling, source apportionment methods
- Statistical Methods: Generalized linear modeling, time series analysis, randomized sample designs, parameter estimation (Bootstrap, Monte Carlo), database development

SELECTED RESEARCH EXPERIENCE

University of Alaska, Fairbanks, *Postdoctoral Research Fellow*, Fairbanks, AK — September 2025–Present
- Organized, executed, and led a time-sensitive field campaign to capture aerial images of high latitude lakes during freeze up (Fall 2025) using drones (Fairbanks and Toolik Lake, AK).
- Independently developed methodology for geospatial analysis at the regional scale.
- Using aerial images of frozen methane bubbles in high latitude lakes and machine learning to improve Arctic carbon balance estimates.

University of California, Irvine, *Graduate Research Scientist*, Irvine, CA — October 2020–June 2025
- Quantified carbon stocks (soils, biomass) and concentrations (biogenic volatile organic compounds, total suspended particles), and transport among Earth systems with vegetation change and permafrost thaw in the Arctic, directly

informing greenhouse gas inventories and methodologies for tracking GHG emissions and natural carbon sequestration

- Led comprehensive field sampling campaigns in remote tundra ecosystems, deploying air monitoring equipment and implementing statistically robust sampling designs and quality control protocols that optimized for significant results while managing logistical constraints
- Analyzed air quality-climate interactions through radiocarbon source apportionment and atmospheric chemistry techniques to understand volatile organic compound emissions from vegetation and secondary organic aerosol formation and primary aerosol emission from boreal forest fires under climate change
- Published 7 peer-reviewed papers (2 first-authored, 2 more in prep) in high-impact journals including *Nature Communications* and *PNAS*, and presented at many conferences (including 3 national-level), demonstrating an ability to translate complex geospatial analyses into actionable scientific insights
- Led cross-functional research teams and mentored undergraduate researchers in carbon measurement techniques, spatial data analysis, and statistical modeling while managing projects

Woodwell Climate Research Center, *Research Scientist Associate*, Falmouth, MA July 2024–December 2024

- Parameterized and ran the Geophysical Institute Permafrost Model (GIPL) for permafrost simulation, implementing spatial statistical methods for ground temperature and greenhouse gas emission prediction
- Applied advanced statistical techniques in a sensitivity analysis for uncertainty quantification using Python
- Secured $50,000 NSF funding to support collaborative research with multidisciplinary scientists, demonstrating ability to identify and translate scientific gaps into funded research outcomes

LEADERSHIP AND INVOLVEMENT

- Board Member, Department of Earth System Science Graduate Leadership
- Seminar Lead, Department of Earth System Science Seminar Series; Half-Baked Seminar Series
- Writer, Administrator, and Grader, Orange County Science Olympiad (Division C)
- Mentor, Earth System Science Undergraduate Research (Undergraduate students: Nicole Nielsen, Yuan Liu)
- Educator, Climate, Literacy, Education, And iNquiry (CLEAN) Initiative
- Mentor and Judge, Irvine Unified School District Science Fair, Orange County Science and Engineering Fair

SELECTED PUBLICATIONS (4 of 7)

Welch, A. M., et al. (2025). Summertime carbonaceous aerosol in interior vs. coastal Northern Alaska. *Journal of Geophysical Research: Atmospheres* https://doi.org/10.1029/2024JD042080

Wang, H., Nagalingam, S., **Welch, A. M.,** Leong, C., Czimczik, C. I., & Guenther, A. B. (2024). Heat waves may trigger unexpected surge in aerosol and ozone precursor emissions from sedges in urban landscapes. *Proceedings of the National Academy of Sciences, 121*(45), e2412817121. https://doi.org/10.1073/pnas.2412817121

Wang, H., **Welch, A. M.**, et al. (2024). High temperature sensitivity of Arctic isoprene emissions explained by sedges. *Nature Communications.* https://doi.org/10.1038/s41467-024-49960-0

Welch, A. M., Pedron, S. A., Jespersen, R. G., Xu, X., Martinez, B., Khazindar, Y., Fiore, N. M., Goulden, M. L., & Czimczik, C. I. (2023). Implications of alder shrub growth for alpine tundra soil properties in Interior Alaska. *Arctic, Antarctic, and Alpine Research, 55*(1). https://doi.org/10.1080/15230430.2023.2285334

SELECTED PRESENTATIONS (3 of 14)

Welch, A., Matthews, T., Wang, H., Leong, C., Nielsen, N., Xu, X., Barsanti, K., Sheesley, R., Guenther, A., Czimczik, C. (2023, December). *Characterization of summertime carbonaceous aerosol and biogenic precursors on the North Slope of Alaska.* Oral presentation at the annual American Geophysical Union Meeting, San Francisco, CA, USA.

Welch, A. M., Nielsen, N., Leong, C., Liu, Y., Wang, H., Teber, D., Guenther, A. B., Bret-Harte, S., Jafarov, E., Czimczik, C. I. (2024, December). *Shrub expansion: Impacts on organic soil and permafrost stability on the North Slope of Alaska*. Poster presented at the annual American Geophysical Union Meeting, Washington, D. C., USA.

Welch, A., Pedron, S., Jespersen, R., Khazindar, Y., Fiore, N., Goulden, M., Martinez, B., Xu, X., Czimczik, C. (2021, December). *Implications of alder shrub expansion on soil properties in Arctic alpine tundra.* Poster presented at the annual fall meeting of the American Geophysical Union, New Orleans, LA, USA.